

$termvar, x$	term variable		
$typvar, X$	type variable		
t	$::=$		term
	x		variable
	$\lambda x. t$	bind x in t	abstraction
	$t t'$		application
	(t)	M	
	$\{t/x\}t'$	M	
v	$::=$		value
	$\lambda x. t$		
T	$::=$		type
	X		variable
	$T \rightarrow T'$		function
	(T)	M	
Γ	$::=$		type environment
	empty		
	$\Gamma, x : T$		
$terminals$	$::=$		
	λ		
	$\longrightarrow$		
	$\rightarrow$		
	$\vdash$		
	$\in$		
$formula$	$::=$		
	<i>judgement</i>		
	not (<i>formula</i>)		
	$x = x'$		
	$x : T \in \Gamma$		
$Jtype$	$::=$		
	$\Gamma \vdash t : T$		
Jop	$::=$		
	$t_1 \longrightarrow t_2$	t_1 reduces to t_2	
$judgement$	$::=$		
	<i>Jtype</i>		
	<i>Jop</i>		
$user_syntax$	$::=$		
	<i>termvar</i>		
	<i>typvar</i>		
	t		
	v		
	T		
	Γ		

| *terminals*
| *formula*

$\boxed{\Gamma \vdash t : T}$

$$\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \quad \text{GTT_VALUE_NAME}$$

$$\frac{\Gamma \vdash t : T_1 \rightarrow T_2 \quad \Gamma \vdash t' : T_1}{\Gamma \vdash t t' : T_2} \quad \text{GTT_APPLY}$$

$$\frac{\Gamma, x_1 : T_1 \vdash t : T}{\Gamma \vdash \lambda x_1. t : T_1 \rightarrow T} \quad \text{GTT_LAMBDA}$$

$\boxed{t_1 \longrightarrow t_2}$ t_1 reduces to t_2

$$\overline{(\lambda x. t_{12}) v_2 \longrightarrow \{v_2/x\} t_{12}} \quad \text{AX_APP}$$

$$\frac{t_1 \longrightarrow t'_1}{t_1 t \longrightarrow t'_1 t} \quad \text{CTX_APP_FUN}$$

$$\frac{t_1 \longrightarrow t'_1}{v t_1 \longrightarrow v t'_1} \quad \text{CTX_APP_ARG}$$

Definition rules: 6 good 0 bad
Definition rule clauses: 12 good 0 bad